12TH BIOLOGY · CHAPTER 18 · CHAPTER TEST

Inheritance

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Biology	**Total Marks:** 26
Chapter: 18	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The blood group AB may be given to a person who has blood group:

(a) A (b) O
(c) B (d) AB

2. When the activity of one gene is suppressed by a non-allelic gene, it is known as:

(a) Incomplete dominance (b) Epistasis
(c) Complete dominance (d) Multiple allelic effect

3. Colour blindness is more common in males because it is a/an:

(a) X-linked recessive trait (b) X-linked dominant trait
(c) Autosomal trait (d) Y-linked trait

4. The F_2 ratio in incomplete dominance is:

(a) 3:1 (b) 9:3:3:1
(c) 1:2:1 (d) 1:1

5. Which of these characters is controlled by a multiple allelic system?

(a) Rh blood group system (b) ABO blood group system
(c) MNS blood group system (d) All of these

6. In birds, the female is:

(a) XX (b) ZZ
(c) ZW (d) XO

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. What is a testcross?
ii. What is co-dominance? Give an example.
iii. Why are O-negative individuals called universal donors?
iv. What is gene linkage?
v. Differentiate between sex-limited and sex-influenced traits.
vi. Normal parents have an albino child. What is the probability that their next child will be normal?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain Mendel's experiment which led to the law of independent assortment.
(b) Explain Erythroblastosis foetalis in the light of antigen-antibody reaction.

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